

General anesthetic agents induce neurotoxicity through oligodendrocytes in the developing brain.

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General anesthetic agents can impact brain function through interactions with neurons and their effects on glial cells. Oligodendrocytes perform essential roles in the central nervous system, including myelin sheath formation, axonal metabolism, and neuroplasticity regulation. They are particularly vulnerable to the effects of general anesthetic agents resulting in impaired proliferation, differentiation, and apoptosis. Neurologists are increasingly interested in the effects of general anesthetic agents on oligodendrocytes. These agents not only act on the surface receptors of oligodendrocytes to elicit neuroinflammation through modulation of signaling pathways, but also disrupt metabolic processes and alter the expression of genes involved in oligodendrocyte development and function. In this review, we summarize the effects of general anesthetic agents on oligodendrocytes. We anticipate that future research will continue to explore these effects and develop strategies to decrease the incidence of adverse reactions associated with the use of general anesthetic agents.

Hyaluronic acid and its receptor CD44, acting through TMEM2, inhibit morphological differentiation in oligodendroglial cells.

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Hyaluronic acid is a main extracellular matrix component in the central nervous system (CNS), which provides structural support under physical and physiological conditions to maintain cellular homeostasis. However, hyaluronic acid and its degradation products are present within focal demyelinating lesions in multiple sclerosis (MS) patients and autoimmune encephalomyelitis (EAE) mouse models. Differentiated plasma membranes called myelin membranes are generated by oligodendrocytes (also called oligodendroglial cells), which are glial cells that wrap neuronal axons in the CNS. Despite these positive or negative relationships of hyaluronic acid with oligodendroglial cell differentiation and/or myelination, it remains unclear whether and how hyaluronic acid affects oligodendroglial cells. Here, we showed that hyaluronic acid and the cognate receptor CD44 are directly involved in inhibiting morphological differentiation in FBD-102b cells, which are differentiation models of oligodendroglial precursor cells, and primary oligodendroglial precursor cells. Their phenotype changes were supported by decreased oligodendroglial cell differentiation, myelin marker protein expression levels, and Akt kinase phosphorylation levels as a marker kinase. Furthermore, the effects of hyaluronic acid required transmembrane protein 2 (TMEM2), a cell surface hyaluronidase. These results suggest that hyaluronic acid and the CD44 receptor, acting through TMEM2, contribute to inhibiting morphological differentiation of oligodendroglial cells, providing a mechanism underlying cell physiological and possible pathological effects responsible for hyaluronic acid.

Hesperetin, a Citrus Flavonoid, Ameliorates Inflammatory Cytokine-Mediated Inhibition of Oligodendroglial Cell Morphological Differentiation.

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Oligodendrocytes (oligodendroglial cells) are glial cells that wrap neuronal axons with their differentiated

plasma membranes called myelin membranes. In the pathogenesis of inflammatory cytokine-related oligodendroglial cell and myelin diseases such as multiple sclerosis (MS), typical inflammatory cytokines tumor necrosis factor α (TNF α) and interleukin-6 (IL-6) are thought to contribute to the degeneration and/or progression of the degeneration of oligodendroglial cells and, in turn, the degeneration of naked neuronal cells in the central nervous system (CNS) tissues. Despite the known involvement of these inflammatory cytokines in disease progression, it has remained unclear whether and how TNF α or IL-6 affects the oligodendroglial cells themselves or indirectly. Here we show that TNF α or IL-6 directly inhibits morphological differentiation in FBD-102b cells, which are differentiation models of oligodendroglial cells. Their phenotype changes were supported by the decreased expression levels of oligodendroglial cell differentiation and myelin marker proteins. In addition, TNF α or IL-6 decreased phosphorylation levels of Akt kinase, whose upregulation has been associated with promoting oligodendroglial cell differentiation. Hesperetin, a flavonoid mainly contained in citrus fruit, is known to have neuroprotective effects. Hesperetin might also be able to resolve pre-illness conditions, including the irregular secretion of cytokines, through diet. Notably, the addition of hesperetin into cells recovered TNF α - or IL-6-induced inhibition of differentiation, as supported by increased levels of marker protein expression and phosphorylation of Akt kinase. These results suggest that TNF α or IL-6 itself contributes to the inhibitory effects on the morphological differentiation of oligodendroglial cells, possibly providing information not only on their underlying pathological effects but also on flavonoids with potential therapeutic effects at the molecular and cellular levels.

Hearing Loss: Genetic Testing, Current Advances and the Situation in Latin America.

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Congenital hearing loss is the most common birth defect, estimated to affect 2-3 in every 1000 births, with ~50-60% of those related to genetic causes. Technological advances enabled the identification of hundreds of genes related to hearing loss (HL), with important implications for patients, their families, and the community. Despite these advances, in Latin America, the population with hearing loss remains underdiagnosed, with most studies focusing on a single locus encompassing the GJB2/GJB6 genes. Here we discuss how current and emerging genetic knowledge has the potential to alter the approach to diagnosis and management of hearing loss, which is the current situation in Latin America, and the barriers that still need to be overcome.

Various aspects of hearing loss in newborns: A narrative review.

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Hearing loss is considered the most common birth defect. The estimated prevalence of moderate and severe hearing loss in a normal newborn is 0.1%-0.3%, while the prevalence is 2%-4% in newborns admitted to the newborn intensive care unit. Neonatal hearing loss can be congenital (syndromic or non-syndromic) or acquired such as ototoxicity. In addition, the types of hearing loss can be conductive, sensorineural, or mixed. Hearing is vital for the acquisition of language and learning. Therefore, early detection and prompt treatment are of utmost importance in preventing the unwanted sequel of hearing loss. The hearing screening program is mandatory in many nations, especially for high-risk newborns. An automated auditory brainstem response test is used as a screening tool in newborns admitted to the newborn intensive care unit. Moreover, genetic testing and screening for cytomegalovirus in newborns

are essential in identifying the cause of hearing loss, particularly, mild and delayed onset types of hearing loss. We aimed to update the knowledge on the various aspects of hearing loss in newborns with regard to the epidemiology, risk factors, causes, screening program, investigations, and different modalities of treatment.

Congenital Deafness and Deaf-Mutism: A Historical Perspective.

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Hearing loss is the most common sensory deficit and one of the most common congenital abnormalities. The estimated prevalence of moderate and severe hearing loss in a normal newborn is 0.1-0.3%, while the prevalence is 2-4% in newborns admitted to the newborn intensive care unit. Therefore, early detection and prompt treatment are of utmost importance in preventing the unwanted sequel of hearing loss on normal language development. The problem of congenital deafness is today addressed on the one hand with hearing screening at birth, on the other with the early (at around 3 months of age) application of hearing aids or, in case of lack of benefit, by the cochlear implant. Molecular genetics, antibody tests for some viruses, and diagnostic imaging have largely contributed to an effective etiological classification. A correct diagnosis and timely fitting of hearing aids or cochlear implants is useful for deaf children. The association between congenital deafness and "mutism", with all the consequences on/the consideration that deaf mutes have had since ancient times, not only from a social point of view but also from a legislative point of view, continued until the end of the nineteenth century, with the development on one side of new methods for the rehabilitation of language and on the other of sign language. But we need to get to the last decades of the last century to have, on the one hand, the diffusion of "universal newborn hearing screening", the discovery of the genetic causes of over half of congenital deafness, and on the other hand the cochlear implants that have allowed thousands of children born deaf the development of normal speech. Below, we will analyze the evolution of the problem between deafness and deaf-mutism over the centuries, with particular attention to the nineteenth century.

Congenital deafness and vestibular disorders: a systematic literature review.

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Congenital deafness is a pathological entity that represents an economical and social burden, affecting up to 0.2% of newborns in Europe. Sensorineural hearing loss (SHL) is caused by a variety of factors, including congenital abnormalities, perinatal infectious diseases and genetic syndromes. The inner ear's vestibular system, nestled alongside the auditory organs, is crucial for balance maintenance. Its close connection with the auditory system means that disturbances in one often coincide with disturbances in the other, highlighting their intertwined functions. With this review we aim to describe objective vestibular tests found in literature and to study their use for diagnosis of vestibular disturbances in patients affected by congenital deafness. The review is conducted with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) 2020 guidelines. The search string used was: [(congenital deafness) OR (congenital hearing loss) OR (congenital hypoacusia)] AND [(vestibular disorders) OR (vertigo)]. An initial abstract reading selection was made, and a subsequent comprehensive full-text reading. For each article, we identified the type of vestibular test utilized and its corresponding outcome. Out of the initial-papers identified through the search string-articles met the

eligibility criteria for further analysis through abstract and full-text reading. After further selection-articles were chosen for detailed examination, focusing on the data of patients. Congenital hearing loss profoundly affects a child's development, especially in language and communication skills, and it is frequently associated with a pathological vestibular system. Early identification allows timely intervention with personalized therapies. In current literature, there is still no gold standard test to identify balance disorders in patients with congenital hearing loss. There is considerable variability on the subject due to the inclusion of diverse patients with various diagnoses, alongside a wide range of available technologies. Managing such conditions necessitates collaboration among healthcare providers, ensuring comprehensive care through prompt diagnosis and personalized treatment plans. Ongoing research aims to further improve screening methods and develop precision medicine approaches tailored to individual needs.

Newborn cytomegalovirus screening: is this the new standard?

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Congenital cytomegalovirus infection (cCMV) is a major cause of childhood hearing loss and neurodevelopmental delay. Early identification of cCMV allows for interventions that improve outcomes, particularly for cCMV-related hearing loss that develops in early childhood. Most cCMV is asymptomatic at birth and is rarely diagnosed without newborn screening. Therefore, various approaches to cCMV screening are increasingly being adopted. Both universal screening (testing all newborns) and targeted screening (testing triggered by failed hearing screening) for cCMV appear valuable, feasible and cost-effective, though universal screening is predicted to have greatest potential overall benefits. CMV PCR testing of newborn oral swabs is sensitive and practical and is therefore widely used in targeted screening programs. In contrast, PCR using dried-blood spots (DBS) is less sensitive but was adopted by current universal cCMV screening initiatives because DBS are already collected from all newborns in high-income countries, which circumvents large-scale oral swab collection. Targeted screening is widely recommended as standard of care, while universal screening is less common but is progressively considered as the optimal strategy for identification of children with cCMV. As with all screening programs, cCMV screening requires commitments to equitable and reliable testing, follow-up and services.

Hearing outcome in congenitally CMV infected children in Finland - Results from follow-up after three years age.

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Cytomegalovirus (CMV) is the most common congenital infection affecting about 0.6% of all newborns in developed countries. Vertical transmission to fetus can take place either after maternal primary or non-primary CMV infection during pregnancy. It is the most common infectious agent for sensorineural hearing loss (SNHL) in young children. The hearing loss after congenital CMV (cCMV) may be present at birth, or may develop after months or even years. In this study, we evaluated hearing outcome at 3-4 years of age in children (n 32) with cCMV identified in universal saliva CMV-PCR-based screening. Study population consisted of mainly asymptomatic children (median age 3.1 years) with cCMV identified in newborn CMV screening. The type of maternal CMV infection (primary or non-primary) was determined

by analyzing CMV antibodies (IgM, IgG and IgG avidity) from preserved maternal serum samples drawn in the end of first trimester of pregnancy. Hearing was evaluated with pure tone audiometry (PTA), or transient-evoked otoacoustic emission (TEOAE) and sound field audiometry (SF). Unilateral hearing loss occurred in 5/32 (16%) of the children with cCMV. None of the subjects in our cohort had bilateral hearing loss. Hearing loss occurred in 3/15 (20%) of children who were born to mothers with non-primary CMV infection during pregnancy, and in 2/10 (20%) of children whose mother had had a primary CMV infection during the 2-3 trimester. None of the additional 6 children, whose mother had primary infection in the first trimester, had hearing loss by age of 3-4 years. Two children with normal hearing at 1 years age had developed unilateral hearing loss by the age of three. Unilateral hearing loss was relatively common among the mainly asymptomatic children with cCMV identified in screening. Long-term follow up of children with cCMV is essential to identify the children with late-onset hearing loss.

Bilateral deafness due to congenital CMV and not rubella reinfection.

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We report on an infant with bilateral deafness discovered at the age of 5 months caused by a retrospectively diagnosed primary maternal CMV infection after definitive exclusion of maternal rubella reinfection as a cause of fetal infection.
